

Solvency Field Theory V12.5

The Quantum Bridge: From Settlement Dynamics
to Born-Compatible Completion Mechanics

A checkpoint architecture from locked-history force response
to Dent/Born pushforward

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June 2026

A retarded field dents the clock plane; the c-budget caps how much of that dent can become real deviation; and the outcome density is the pushforward of canonical completion cells through that bounded map.

Publication boundary. This document consolidates a continuous candidate architecture from force response to Born-compatible completion dynamics. It claims structural bridges, not a completed derivation of quantum mechanics. The provenance is the R1–R60 FORCE audit sequence. Every claim is classified as earned, conditional, or open.

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1 The physical picture

Every quantum mechanics textbook begins with axioms. The state is a vector in Hilbert space. Observables are Hermitian operators. Measurement outcomes are eigenvalues. The probability of an outcome is the squared modulus of the amplitude. Time evolution is unitary.

These axioms work. Quantum mechanics is the most precisely tested theory in the history of physics. Nothing in this document disputes the predictions. What this document asks is: *why these axioms?* Not “are they correct” — they are — but “what is it about the structure of reality that makes these the right rules?”

The standard answer is silence. The axioms are postulated because they reproduce experiment. No deeper justification is offered. The Born rule — probability equals amplitude squared — is especially opaque. Why *squared*? Why not the amplitude itself, or the cube, or something else entirely? The axiom is a door that opens onto all of quantum physics, and nobody knows why that particular door exists.

This document proposes that the door exists because of the geometry of a clock.

1.1 The clock hand and its plane

A massive particle in SFT cycles internally at its Compton frequency, $\omega_C = mc^2/\hbar$. This is not a metaphor. It is the measured rest energy expressed as a rate through Planck’s relation. The particle is a clock. Its internal state at any moment is described by a phase — an angle on a circle, sweeping continuously at ω_C .

A phase needs two numbers to specify. Not one. An angle is not a point on a line; it is a direction in a plane. To represent where the clock hand points without losing the direction, you need two coordinates: $x = r \cos \phi$ and $y = r \sin \phi$. The compact notation for these two coordinates is a complex number, $a = x + iy = re^{i\phi}$.

The complex amplitude of quantum mechanics is not an abstract mathematical device imported for calculational convenience. It is the natural coordinate system for a phase-bearing clock. Two quadratures. One plane. One complex number.

And the *size* of an unresolved possibility on that plane is not the length of the clock hand. It is the *area* swept by the clock hand in its plane. Length gives $|a|$. Area gives $|a|^2$. The Born rule — probability proportional to $|a|^2$ — is the statement that the natural measure of a phase-plane possibility is its area, not its radius.

1.2 Why the square is forced

This is not an assertion. It was tested. Among all candidate measures $\mu(a) = |a|^p$ for $p = 1, 2, 3, \dots$, only $p = 2$ survives three independent consistency requirements:

Rotation invariance. Reversible mixing of two possibility components — rotating the clock hand by an angle θ between them — must not create or destroy total measure. The identity $\cos^2 \theta + \sin^2 \theta = 1$ is the *only* trigonometric identity that holds for all angles. Any other power p changes under rotation. The square is selected by the geometry of the circle.

Fine-graining invariance. Splitting one possibility into n equal sub-possibilities should not change the total measure. For a conserved quadratic budget $\sum |a_i|^2 = N$, equal splitting sends $a \rightarrow a/\sqrt{n}$, giving $n \times |a/\sqrt{n}|^2 = |a|^2$. The square root is not quantum mechanical magic. It is the price of conserving the square.

Canonical area. In action-angle coordinates (J, ϕ) , the natural phase-space measure is $dJ d\phi$. Hamiltonian flow preserves this measure (Liouville’s theorem). Completion cells that are equal in

$dJ d\phi$ remain equal under any smooth dynamics. The accumulated area of branch k is $\mu_k = 2\pi I_k = \pi|a_k|^2$, recovering the Born weight after normalization.

Three independent arguments. Same answer. Probability is amplitude squared because the natural size of a clock-plane possibility is its area, and area is the only measure that all three consistency conditions preserve.

1.3 Probability as a mechanical pushforward

With the measure established, the question becomes: what determines the *distribution* of outcomes in a specific experiment?

Consider a particle approaching a double slit. It has a definite trajectory, a definite internal phase, and a definite c-budget partition. It interacts with the slit atoms, which place phase-dependent forces — “dents” — on its trajectory. The deflection depends on where in its Compton cycle the particle happens to be at the moment of transit. Different phases produce different deflections. Fire many particles, each arriving at a random phase, and the distribution of deflections on the detector screen is the pushforward of the uniform phase measure through the phase-dependent interaction geometry.

This gives a five-input recipe for any quantum outcome:

1. **Anchor:** the current trajectory, set by the written history. The baseline path the particle follows if nothing acts on it.
2. **Species:** the particle type and mass, setting the Compton frequency and the c-budget partition between internal and external motion.
3. **Velocity:** the current spatial speed, determining how much c-budget remains for internal cycling and therefore the phase density.
4. **Phase:** the current position in the Compton cycle. Unknown to the experimenter. The source of all “quantum randomness.”
5. **Dents:** every external interaction — electromagnetic fields, gravitational quadrupoles, slit geometry — placing phase-dependent forces on the trajectory.

The outcome density on the detector is the Jacobian of the phase-to-deviation map:

$$P(x) = \sum_{\phi: f(\phi)=x} \frac{1}{2\pi} \left| \frac{d\phi}{dx} \right|. \quad (1)$$

For the full canonical measure, the pushforward is:

$$P(x) = \int_{\Omega} \delta(x - f(\omega)) dJ d\phi, \quad (2)$$

where $\omega = (J, \phi, \text{history, fields, geometry, c-budget})$ ranges over all lawful completions and f maps each completion to an outcome through the retarded, bounded interaction geometry.

This is not a probability postulate. It is a mechanical computation. The particle is real. Its trajectory is definite. Its phase is definite but unknown. The “quantum randomness” is the experimenter’s ignorance of the internal phase, not an intrinsic indeterminacy of nature. The interference pattern is the structure of the Jacobian, arising from the periodic phase dependence of the dent interaction. And the Born rule — $P(k) = |a_k|^2 / \sum |a_j|^2$ — is the canonical area share, which equals the Jacobian integral over the appropriate phase-plane domain.

1.4 Why the Schrödinger equation

If the particle maintains an internal action phase S , with $E = -\partial_t S$ and $\mathbf{p} = \nabla S$, then writing $\Psi = R e^{iS/\hbar}$ immediately gives the Schrödinger generators:

$$i\hbar \partial_t \Psi = E \Psi, \quad -i\hbar \nabla \Psi = \mathbf{p} \Psi. \quad (3)$$

In the nonrelativistic limit with $E - mc^2 \approx p^2/(2m) + V$, this becomes:

$$i\hbar \partial_t \psi = \left(-\frac{\hbar^2}{2m} \nabla^2 + V \right) \psi. \quad (4)$$

The Schrödinger equation is not a separate axiom. It is the bookkeeping of how the action clock changes in time and space. The equation is the house frame; the foundation is the clock.

1.5 When possibilities become facts

Before a write event, the particle's future is a superposition of possibilities — the pre-write alternatives that can interfere, add, and form patterns. After the write, one possibility has been inscribed on the boundary. The write is irreversible (no-overwrite). The outcome is idempotent: $b^2 = b$. Once written, it is a fact, not a possibility.

Decoherence is the process that makes branches *countable* — it creates records that distinguish one possibility from another. But decoherence does not select which branch gets written. It prepares the branches for settlement. Settlement — the actual inscription on the boundary — is the separate, irreversible step that turns a countable branch into a written fact.

Before the write, possibilities can add. After the write, facts are idempotent.

This separation resolves the measurement problem without invoking collapse, consciousness, or many-worlds branching. Decoherence creates the options. Settlement chooses one. The Born rule determines the probability of each option through the canonical area of its completion domain. No new physics is invoked. The same no-overwrite rule that gives time its arrow and expansion its direction also gives measurement its irreversibility.

1.6 The architecture that follows

With these physical pictures in hand, the rest of this document is their precise mathematical articulation, compressed from sixty audited gates.

Section 2 covers the force-to-settlement architecture: locked-history stiffness, charge as holonomy, branch sockets, and idempotent settlement (R1–R30). Section 3 derives the Born bridge: complex amplitudes, rotation/fine-graining/canonical-area selection of $|a|^2$, and the action-angle measure (R31–R36, R50–R54). Section 4 establishes the Schrödinger bridge: action phase, generators, free kinetics, and interaction structure (R37–R42). Section 5 separates decoherence from settlement and defines the completion domain (R43–R50). Section 6 develops the Dent/Born push-forward: retarded potential, c-budget cap, phase-to-deviation map, and response functions (R51–R60). Section 7 presents the full claim boundary. The gate rollup is in the appendix.

2 From forces to settlement

2.1 Locked-history stiffness

A worldline is written history. Each tick inscribes one Planck area on the boundary. The past is consumed and cannot be revisited. This rigidity constrains the next write: any deviation from the current trajectory must reconcile with the locked history.

A force is therefore not a push through a background medium. It is a constrained reallocation of the next write:

A force is the local observable response of real one-speed structures reallocating causal budget under compatibility and settlement constraints.

The species-specific compliance — how readily a given particle deviates — is set by its c-budget partition. Heavy particles resist deviation because most of their budget is locked into fast internal cycling. The maximum lateral deviation per tick is:

$$\Delta_{\perp, \max} = \frac{c}{\gamma} \Delta t. \quad (5)$$

2.2 Charge as holonomy carrier

The FORCE audit (R9–R18) established that electromagnetic charge cannot be the Compton phase, cannot be chirality alone, and must be a separate discrete holonomy carrier on the worldline. Continuous direction participation is always positive:

$$p_i = \frac{\delta_i^2}{\delta_1^2 + \delta_2^2 + \delta_3^2}, \quad p_i \geq 0, \quad \sum_i p_i = 1. \quad (6)$$

Charge sign is a separate orientation or holonomy property, not a negative participation share.

For fractional charges, the minimal non-self-conjugate residue system is $N = 3$: halves are their own conjugates ($1 = -1 \pmod{2}$), but thirds have distinct partners ($1 \neq -1 \pmod{3}$). If charge holonomy requires distinct conjugates, $N = 3$ is selected as the minimum, yielding $+2/3$ and $-1/3$ as the fractional charge labels inside closure-complete bundles.

2.3 Branch sockets

The branch structure (R19–R26) models weak-force-like transitions as two-state sockets, not infinite ladders:

$$B_+|L\rangle = |U\rangle, \quad B_+|U\rangle = 0, \quad B_-|U\rangle = |L\rangle, \quad B_-|L\rangle = 0. \quad (7)$$

Nilpotency ($B_+^2 = B_-^2 = 0$) follows from one-write-per-tick: a branch socket has two statuses (not-yet-switched, switched-once), and a second same-direction switch would require rewriting the same socket, which no-overwrite forbids.

2.4 Idempotent settlement

A written branch status is idempotent: $b^2 = b$. The pre-settlement state is a superposition $\psi = a_L|L\rangle + a_U|U\rangle$. Settlement maps this to a definite outcome:

$$S(\psi, P, E_1, \dots) \rightarrow b \in \{0, 1\}, \quad b^2 = b. \quad (8)$$

At this stage the Born weighting is compatible but not yet derived.

3 The Born bridge

3.1 Complex amplitudes from phase structure

A phase-bearing possibility needs two real quadratures: $a = x + iy = re^{i\phi}$ (R36). The natural invariant measure on the two-dimensional amplitude plane is $d\mu = dx dy$.

3.2 Rotation selection

Among candidate budgets $\sum |a_i|^p$, only $p = 2$ is invariant under arbitrary rotations of possibility components (R34):

$$|\cos \theta|^p + |\sin \theta|^p = 1 \quad \forall \theta \quad \Longleftrightarrow \quad p = 2. \quad (9)$$

3.3 Fine-graining selection

If a branch of amplitude a is split into n equal subbranches with amplitude $a/\sqrt{n}$ (required by quadratic norm conservation), then a candidate measure $\mu \sim |a|^q$ satisfies (R32):

$$|a|^q = n \left| \frac{a}{\sqrt{n}} \right|^q = |a|^q n^{1-q/2} \quad \Longrightarrow \quad q = 2. \quad (10)$$

3.4 Action-angle canonical measure

Writing the amplitude in action-angle form (R52):

$$a = \sqrt{2I} e^{i\phi}, \quad |a|^2 = 2I, \quad dx dy = dI d\phi. \quad (11)$$

With physical action coordinate $J = \hbar I$:

$$dJ d\phi = \hbar dI d\phi. \quad (12)$$

For canonical flow $\dot{\phi} = \partial H / \partial J$, $\dot{J} = -\partial H / \partial \phi$, the divergence vanishes identically (Liouville):

$$\frac{\partial \dot{J}}{\partial J} + \frac{\partial \dot{\phi}}{\partial \phi} = -\frac{\partial^2 H}{\partial J \partial \phi} + \frac{\partial^2 H}{\partial \phi \partial J} = 0. \quad (13)$$

Equal canonical cells remain equal under any Hamiltonian evolution. The Born weight is the canonical area share:

$$P(k) = \frac{|a_k|^2}{\sum_j |a_j|^2}. \quad (14)$$

The physical interpretation: S is the odometer (accumulated action). ϕ is the clock hand (current phase). I is the completion capacity (how much clock-plane territory is available for the next write). $J = \hbar I$ is that territory in action units. Born weight is the share of canonical territory, not the particle's total action.

4 The Schrödinger bridge

4.1 Action phase and generators

With $\Psi = R e^{iS/\hbar}$, $E = -\partial_t S$, and $\mathbf{p} = \nabla S$ (R37):

$$i\hbar \partial_t \Psi = E \Psi, \quad -i\hbar \nabla \Psi = \mathbf{p} \Psi. \quad (15)$$

4.2 Free worldline kinetics

The free action is $dS_{\text{free}} = -mc^2 d\tau$ (R38–R39). The Hamiltonian is $H = \gamma mc^2$, and in the nonrelativistic branch:

$$H - mc^2 \approx \frac{p^2}{2m}, \quad (16)$$

giving the standard kinetic Schrödinger operator $-\hbar^2 \nabla^2 / (2m) + V$.

4.3 Interaction structure

Interactions enter the action as one-forms (R40): $S_{\text{int}} = \int B_\mu dx^\mu$. The gauge-invariant field strength is $F_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu$ (R42), with stiffness action $\sim \int F_{\mu\nu} F^{\mu\nu} d^4x$ and source coupling $\int B_\mu J^\mu d^4x$, where $\partial_\mu J^\mu = 0$.

Gauge relabeling invariance (R42) says: charge curvature $F = dB$, not the connection label B itself. The field energy is proportional to curvature squared — the minimal cost of bending the connection.

5 Decoherence, settlement, and admissible completions

5.1 Amplitude curvature and quantum potential

The Madelung decomposition $\Psi = R e^{iS/\hbar}$ produces a quantum potential (R43):

$$Q = -\frac{\hbar^2}{2m} \frac{\nabla^2 R}{R}, \quad (17)$$

which provides the non-classical force that distinguishes quantum trajectories from classical ones. The coefficient, uniqueness, and node protocol remain open.

5.2 Linearity before settlement

Pre-write amplitude evolution is governed by a quadratic action (R44):

$$\mathcal{A}[\psi] = \int \left[\frac{i\hbar}{2} (\psi^* \partial_t \psi - \psi \partial_t \psi^*) - \frac{\hbar^2}{2m} |\nabla \psi|^2 - V |\psi|^2 \right] d^3x dt. \quad (18)$$

Linearity holds before settlement because the quadratic action produces linear Euler–Lagrange equations. After settlement, the outcome is idempotent: linearity is broken by the write.

Before the write, possibilities can add. After the write, facts are idempotent.

5.3 Decoherence creates records, not outcomes

Branch-record entanglement (R45–R47):

$$\left(\sum_i a_i |i\rangle \right) |E_0\rangle \rightarrow \sum_i a_i |i\rangle |E_i\rangle. \quad (19)$$

The reduced coherence is $\rho_{ij} = a_i a_j^* \langle E_j | E_i \rangle$. As the environment records accumulate, the off-diagonal terms $\langle E_j | E_i \rangle \rightarrow 0$, making branches countable. But countable is not settled. Decoherence prepares the menu. Settlement orders.

5.4 Admissible completions

The lawful completion space is (R48–R50):

$$\Omega = \bigsqcup_k \Omega_k, \quad \Pr(k) = \frac{\mu(\Omega_k)}{\sum_j \mu(\Omega_j)}. \quad (20)$$

The non-smuggling route to the Born measure is the quadrature-area completion (R50):

$$\mu(\Omega_k) \propto \pi |a_k|^2. \quad (21)$$

This is earned as a structural candidate. The physical theorem proving these area cells correspond to real ledger completions remains open.

6 The Dent/Born pushforward

6.1 Pre-write Hamiltonian

The pre-write dynamics lives on the canonical action-angle plane (R55):

$$H_{\text{pre}}(J, \phi; t) = \Omega(t) J + U(J, \phi; \text{external dents}), \quad (22)$$

where Ω is the Compton cadence (R56):

$$\Omega_\tau = \omega_C = \frac{mc^2}{\hbar}, \quad (23)$$

derived from the free worldline action $dS = -mc^2 d\tau$, not guessed.

6.2 Retarded dent potential

External interactions place phase-dependent “dents” on the trajectory through a retarded potential (R57):

$$U_{\text{dent}} = \epsilon_{\text{ret}}(x, t) J F(\phi - \chi_{\text{ret}}(x, t)), \quad (24)$$

where ϵ_{ret} and χ_{ret} are the amplitude and orientation of the dent, determined by the past-light-cone source history. The source law governing these quantities remains open.

6.3 Maximum deviation cap

The c-budget constrains the maximum lateral deviation per tick (R58):

$$\Delta_{\perp, \text{max}} = \frac{c}{\gamma} \Delta t. \quad (25)$$

This is not a new force constant. It is the edge of the next-write sandbox, set by the same one-speed budget that determines mass and time dilation.

6.4 Projection and response

The accumulated dent demand is projected into bounded lateral deviation (R59):

$$\delta v_{\perp} = \frac{c}{\gamma} \tanh(\eta_{\text{dent}}), \quad (26)$$

where η_{dent} is the integrated dent-impulse. In the weak-dent limit, $\delta v_{\perp} \approx (c/\gamma)\eta_{\text{dent}}$. In the strong-dent limit, $|\delta v_{\perp}| \rightarrow c/\gamma$ — the budget is saturated. The response functions (R60) are:

$$G_{\phi}(\theta) = \cos \theta, \quad G_{\perp}(\theta) = \sin \theta, \quad (27)$$

where $\theta = \phi - \chi_{\text{ret}}$ is the phase offset between the particle's clock hand and the dent's orientation.

6.5 The pushforward

The observed outcome density is the pushforward of the canonical completion measure through the bounded interaction map:

$$P(x) = \int_{\Omega} \delta(x - f(\omega)) dJ d\phi. \quad (28)$$

Branch probabilities are:

$$P(k) = \frac{\mu(\Omega_k)}{\mu(\Omega)}. \quad (29)$$

The final settlement is idempotent: $b_k = 1$, $b_{j \neq k} = 0$, $b_i^2 = b_i$.

7 What is earned, conditional, and open

7.1 Earned

A continuous candidate architecture from force response to Dent/Born completion.
A force-as-next-write-response picture constrained by locked history.
Charge as holonomy/address carrier, separated from participation shares.
Branch sockets and idempotent settlement variables.
Action phase and generator bridge into Schrödinger structure.
Decoherence as branch-record formation, not selection.
Three independent routes to $|a|^2$ as the Born measure: rotation invariance, fine-graining invariance, and canonical area.
Action-angle canonical measure preserved by Liouville flow.
Retarded Dent mechanics with c-budget cap and response constraints.

7.2 Conditional

Born recovery if the physical completion domains carry canonical area measure.
Dent/Born envelope recovery if retarded source and projection maps produce the required outcome domains.
Standard effective quantum behavior if the pushforward measure and selector preserve Born marginals.

7.3 Open

Retarded source law ($\epsilon_{\text{ret}}, \chi_{\text{ret}}$).
Projection normalization ($\kappa, \alpha_{\text{proj}}$).
Physical Ω -domain theorem.
Quantitative Dent pushforward toy tests.
Single-branch selection and no-signaling theorem.
 $\hbar$ normalization.
Full constants (α , Standard Model masses, mixing parameters).

7.4 Rejected routes

Charge as Compton phase: rejected. The mass clock is species-dependent; charge is universal.

Born from phase-length alone: rejected. Phase length gives $|a|$, not $|a|^2$. The full action-angle area is required.

Infinite branch ladders: rejected. Physical branch access is a two-state socket truncated by one-write-per-tick nilpotency.

Decoherence as selection: rejected. Decoherence creates countable records. Selection is a separate settlement step.

8 Conclusion

The result of V12.5 is not closure. It is continuity.

SFT now has a preserved candidate architecture from force response to settlement and from settlement to Born-compatible completion measure. The key shift is that probability is no longer introduced as a free ghost rule. It is approached as a measure over lawful unresolved completions, with canonical action-angle area providing the first non-smuggling route to the $|a|^2$ weighting. Retarded dents then supply a mechanical way for history, fields, geometry, and the c-budget to push those completion cells into outcomes.

The open problems are equally explicit: the retarded source law, the projection normalization, quantitative envelope recovery, and the no-signaling selection theorem remain future work. The value of V12.5 is a clean map: what has been structurally earned, what remains conditional, and where the next tests must attack.

The Dent/Born bridge is now a mechanical program, not a metaphor.

The program is not finished, but the map is finally sharp enough to test.

Acknowledgments

The R1–R60 FORCE audit sequence was developed and executed in collaboration with ChatGPT/5.5 (OpenAI). The physical narrative was developed with Claude/Opus (Anthropic). The five-input mechanical recipe for Born probabilities emerged from collaboration between all three. The theoretical framework is SFT [1, 2, 3, 4].

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A Gate rollup

Gate	Title	Verdict
R1–R8	Locked-history stiffness	Pass: force as next-write response; species compliance; field-stiffness separation. Constants open.
R9–R18	Charge holonomy carrier	Pass: continuous participation separated from discrete charge; $N=3$ minimal non-self-conjugate system.
R19–R26	Branch socket algebra	Pass: two-state socket; nilpotent operators from one-write-per-tick. Socket dynamics open.
R27–R30	Idempotent settlement	Pass: $b^2=b$ ledger predicates; settlement map. Born-compatible, not Born-derived.
R31–R34	Born measure selection	Conditional pass: rotation and fine-graining select $p=2$ uniquely. Hilbert-splitting origin open.
R35–R36	Unitary mixing and complex amplitudes	Pass: budget-preserving basis changes are unitary-like; phase-bearing objects need complex coordinates.
R37–R39	Schrödinger generator bridge	Conditional pass: action phase gives generator form; free action gives kinetic operator. $\Lambda \rightarrow S$ open.
R40–R42	Interaction one-form and curvature	Pass: one-form coupling; $F=dB$ gauge invariant; curvature stiffness. Source laws open.
R43	Quantum potential	Conditional: native amplitude-curvature route to Q . Coefficient and uniqueness open.
R44	Linearity before settlement	Pass: quadratic action gives linear pre-write evolution; settlement breaks linearity.
R45–R47	Decoherence / branch records	Pass: decoherence creates countable branches; does not select outcomes.
R48–R50	Admissible completions and area measure	Conditional pass: quadrature-area gives non-smuggling Born candidate. Physical Ω theorem open.
R51	Phase-to-deviation Jacobian	Pass: Born as pushforward of uniform phase through interaction geometry. Full measure needs $dJ d\phi$.
R52–R54	Action-angle canonical measure	Pass: $dx dy = dI d\phi$; Liouville preserves equal cells; Born weight is canonical area share.
R55–R56	Pre-write Hamiltonian	Conditional pass: $H_{\text{pre}} = \Omega J + U$; Compton cadence from free action.
R57	Retarded dent potential	Structural pass: retarded $\epsilon_{\text{ret}}, \chi_{\text{ret}}$ with past-light-cone source data. Source law open.
R58	Maximum deviation cap	Pass: $\Delta_{\perp, \text{max}} = c \Delta t / \gamma$ from c-budget.
R59–R60	Projection and response functions	Pass: bounded tanh projection; $\cos \theta / \sin \theta$ response. Source law and normalization open.